Linux Basic Commands

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1. File & Directory Commands

Command	Full Form	Kaam (Work)	Example
pwd	Print Working Directory	Batata hai aap kahan ho (current folder)	`pwd`
ls	List	Folder ke andar kya kya hai dikhata hai	`ls` `ls -l` (details) `ls -a` (hidden files)
cd	Change Directory	Folder change karna	`cd /home` `cd ..` (peeche jana) `cd ~` (home folder)
mkdir	Make Directory	Naya folder banana	`mkdir newfolder`
rmdir	Remove Directory	Khaali folder delete karna	`rmdir oldfolder`
rm	Remove	File ya folder delete karna	`rm file.txt` `rm -rf foldername` (force delete)
cp	Copy	File/folder copy karna	`cp file1.txt file2.txt` `cp -r folder1 folder2`
mv	Move	File move karna ya rename karna	`mv old.txt new.txt` (rename) `mv file.txt /home/` (move)
touch	-	Khaali file banana	`touch newfile.txt`
find	-	File dhoondna	`find /home -name "*.txt"`

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2. File Viewing & Editing Commands

Command	Kaam	Example
cat	File ka content dekhna	`cat file.txt`
less	File scroll karke dekhna (q press karke exit)	`less bigfile.log`
head	File ke pehle 10 lines dikhana	`head file.txt`
tail	File ke last 10 lines dikhana	`tail file.txt` `tail -f logfile` (live dekhna)
nano	Simple text editor	`nano file.txt`
vi / vim	Advanced editor	`vi file.txt`

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3. Permission & Ownership Commands

> *Concept: Linux mein 3 permissions hote hain: **Read ®, Write (w), Execute (x)***

Command	Kaam	Example
chmod	Permission change karna	`chmod 755 script.sh` `chmod +x file` (executable banana)
chown	Owner change karna	`chown user:group file.txt`
sudo	Admin power se command chalana	`sudo apt update`

Permission Numbers Yaad Rakho:

- **7 = Read + Write + Execute (rwx)**
- **6 = Read + Write (rw-)**
- **5 = Read + Execute (r-x)**
- **4 = Read Only (r–)**

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4. User Management Commands

Command	Kaam	Example

whoami	Batata hai current user kaun hai	`whoami`
id	User ka UID, GID dikhata hai	`id username`
adduser / useradd	Naya user banana	`sudo adduser john`
passwd	Password change karna	`passwd` `sudo passwd john`
usermod	User modify karna (group add karna)	`sudo usermod -aG sudo john`
su	Switch user	`su - root`
who	Kaun kaun logged in hai	`who`

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5. System Information Commands

Command	Kaam	Example
uname -a	Kernel aur OS info	`uname -a`
df -h	Disk space kitna bacha hai	`df -h`
du -sh	Folder ka size kitna hai	`du -sh /home`
free -h	RAM kitni bachi hai	`free -h`
top / htop	Live CPU/RAM usage dekhna	`top` (q se exit)
uptime	System kitne din se chal raha hai	`uptime`
history	Purane commands dekho	`history`

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6. Package Management (Software Install)

> *Ubuntu/Debian vs CentOS/RHEL*

Kaam	Ubuntu / Debian	CentOS / RHEL / Fedora
Update list	`sudo apt update`	`sudo yum update`

Install	`sudo apt install nginx`	`sudo yum install nginx`
Remove	`sudo apt remove nginx`	`sudo yum remove nginx`
Search	`apt search nginx`	`yum search nginx`

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7. Network Commands

Command	Kaam	Example
ip addr / ifconfig	IP address dekhna	`ip a`
ping	Network check karna	`ping google.com`
ssh	Remote server connect karna	`ssh user@192.168.1.10`
scp	File transfer (remote)	`scp file.txt user@ip:/path`
wget / curl	Internet se file download	`wget url`
netstat -tulnp	Open ports dekhna	`netstat -tulnp`
hostname	System ka naam	`hostname`

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8. Process Management

Command	Kaam	Example
ps aux	Sabhi running processes dekho	`ps aux | grep nginx`
kill	Process band karna (by PID)	`kill 1234`
kill -9	Force band karna	`kill -9 1234`
systemctl	Service control (start/stop)	`systemctl start nginx` `systemctl enable nginx`
jobs	Background jobs dekho	`jobs`

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9. Compression & Archive

Command	Kaam	Example
tar	Files ko pack/unpack karna	`tar -czvf backup.tar.gz folder/` (zip) `tar -xzvf backup.tar.gz` (unzip)
gzip / gunzip	.gz files compress	`gzip file.txt`
zip / unzip	.zip files	`zip -r file.zip folder/`

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10. Shortcut Keys (Terminal mein kaam aate hain)

Shortcut	Kaam
Tab	Command/filename auto complete
Ctrl + C	Current command rok do (Cancel)
Ctrl + D	Logout / Exit
Ctrl + L	Screen saaf karo (clear)
Ctrl + R	Purana command search karo
Up/Down Arrow	Previous/Next command
!!	Last command dobara chalao

Interview Important Questions

Q1. `rm` aur `rmdir` mein difference?

> *`rmdir` sirf **khaali folder** delete karta hai. `rm -rf` folder ke saath uske andar ki files bhi delete karta hai.*

Q2. `chmod 777` ka matlab?

> *Sabko (User, Group, Others) **Read, Write, Execute** permission dena. (Security risk hota hai).*

Q3. Process ko background mein kaise chalayein?

> *Command ke last mein `&` laga do. Example: `firefox &`*

Q4. Hidden files kaise dekhein?

> `ls -a` *(Hidden files* `.` *se start hote hain).*

Q5. `sudo` **ka full form?**

> ***Super User DO*** *- Admin rights se command chalana.*

One Page Cheat Sheet

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# Navigation
pwd, ls, cd, cd ..

# Files
touch, cp, mv, rm, mkdir

# View
cat, less, head, tail

# Permissions
chmod, chown, sudo

# System
df -h, free -h, top, uname -a

# Network
ip a, ping, ssh, netstat -tulnp

# Package
apt update && apt install package

# Process
ps aux, kill PID, systemctl start name
```

> ***Tip:*** *Terminal mein* `man command` *type karke kisi bhi command ka manual padh sakte ho. (Example:* `man ls` *)*